KRATOS

Kratos User Manual

Installing and running the agentic TMDD calibration, population-estimation and reporting platform

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Contents

1 What Kratos does

2 Requirements

3 Installation
- 3.1 Windows (PowerShell)
- 3.2 Linux (bash)

4 Quick start: the synthetic demonstration

5 Running your own analysis
- 5.1 Step 1 — Prepare the data file
- 5.2 Step 2 — Write the configuration
- 5.3 Step 3 — Validate the data
- 5.4 Step 4 — Run the workflow
- 5.5 Step 5 — Follow progress, cancel, resume
- 5.6 Step 6 — Read the results

6 Command reference

7 Configuration reference

8 The browser interface

9 What a run directory contains

10 Reports

11 Reproducing the mAb analysis in this project

12 Advanced use

13 Troubleshooting

14 Glossary

1 What Kratos does

Kratos takes a pharmacokinetic dataset (CSV) and an analysis configuration (YAML) and runs the complete target-mediated drug disposition (TMDD) workflow automatically, recording every step:

Stage	What happens	Main output
1. Validate	Maps columns, checks every record, converts units to h / nmol / nmol/L, flags suspicious data (never deletes)	`data/normalized_data.csv`, `data/validation_report.json`
2. Raw plots	Concentration–time plots (linear/log), individual profiles, dose-normalised overlays, timelines	`figures/raw_*.png`
3. Model	Loads the approved, versioned TMDD template and checks parameters, bounds and units	`model/model_spec.json`
4. Calibrate	Bounded multistart fitting (Python/SciPy) guided by an adaptive planner	`calibration/calibration_history.csv`
5. Estimate	nlmixr2 **FOCEi** population estimation in R from the best calibration starts	`estimation/final/fit.rds`
6. Diagnose	Convergence, covariance, RSE, bounds, shrinkage, CWRES, VPC; heuristic review flags	`diagnostics/diagnostics_summary.json`, `review_flags.json`
7. Report	Tables, figures and an HTML/PDF report built only from saved results	`report/moDoc.html`, export zip

> **Important**
>
> Kratos automates the numerical work; it does **not** declare a model scientifically valid. Review flags are aids for the scientist. Changes to model structure, fixed parameters or data exclusions are always explicit decisions written into the configuration.

2 Requirements

Component	Version used and tested	Needed for
Windows 10/11 or Linux	Windows 10 (19045)	everything
Python	3.12 (3.10 or newer)	the whole platform
Python packages	numpy, scipy, pandas, matplotlib, PyYAML, jsonschema (see `requirements-lock.txt`)	core
R + Rtools (Windows)	R 4.5.1 with Rtools45	population estimation, VPC, bootstrap
R packages	rxode2 5.1.6, nlmixr2 4.0.1, nlmixr2est 4.1.0, jsonlite, ggplot2	population estimation
Quarto + Jupyter	Quarto 1.10, jupyter/ipykernel	HTML report
Microsoft Edge or Google Chrome	any recent version	PDF report
streamlit	1.44	browser interface (optional)

Without R, Kratos still runs and produces an explicitly labelled **calibration-only** analysis; population estimation is marked `not_run`.

3 Installation

3.1 Windows (PowerShell)

```
cd C:\Users\devamdesai\Downloads\Kratos

# 1. Python package and tools
python -m pip install -e .
python -m pip install streamlit pytest jupyter ipykernel

# 2. R packages (project library .Rlib is used first; installs anything missing)
$env:R_LIBS = "$PWD\.Rlib"
& "$env:LOCALAPPDATA\Programs\R\R-4.5.1\bin\Rscript.exe" R\setup_packages.R --install

# 3. Check
kratos --help
python -m pytest -q          # about 20 minutes; R tests are skipped automatically if R is
         missing
```

3.2 Linux (bash)

```
cd Kratos
python3 -m pip install -e . streamlit pytest jupyter ipykernel
export R_LIBS="$PWD/.Rlib"
Rscript R/setup_packages.R --install
kratos --help
```

Tip

Kratos finds `Rscript` automatically (PATH, then standard install folders). To point it at a specific R, set `KRATOS_RSCRIPT`. To force the calibration-only path, set `KRATOS_DISABLE_R=1`. Copy `.env.example` to `.env` for these settings; never put credentials in configuration files.

4 Quick start: the synthetic demonstration

This reproduces the verified demonstration, using simulated data with known parameters, in about 15–40 minutes, most of it the FOCEi fit.

```
cd C:\Users\devamdesai\Downloads\Kratos
python scripts\generate_synthetic.py                              # writes data\synthetic\...
kratos run --config config\synthetic_demo.yaml --run-id demo01    # full workflow
start runs\demo01\report\moDoc.html                               # open the report
```

When it finishes, `runs\demo01\status.json` shows every stage as `succeeded`. The report states which analysis mode ran: *population estimation (nlmixr2 FOCEi) + pooled calibration*.

5 Running your own analysis

5.1 Step 1 — Prepare the data file

One row per record, with a header. The column names can be anything; you map them in the configuration.

Field	Required	Meaning
`ID`	yes	subject or profile identifier (kept as is)
`TIME`	yes	time since first dose, ≥ 0
`DV`	yes	observed concentration (empty on dose rows)
`AMT`	dose rows	dose amount (> 0)
`EVID`	optional	0 = observation, 1 = dose. Or set `evid_rule: amt_present` to treat rows with AMT as doses
`ROUTE` / `CMT`	optional	route code mapped to `iv_bolus`, `iv_infusion` or `sc`
`RATE` / `DUR`	optional	infusion rate or duration (both given → must agree)
`ADDL` / `II`	optional	additional repeat doses and interval
`DVID`	optional	endpoint code when more than one assay is present
`CENS` / `LLOQ`	optional	censoring; censored records are currently **rejected** (no LLOQ/2 substitution)

The full rules are in `docs/data_dictionary.md`.

5.2 Step 2 — Write the configuration

Copy an existing file (`config/mab_partA.yaml` or `config/synthetic_demo.yaml`) and edit it. The key parts:

```
project: my_study                    # letters, digits, _ . - only
data:
  path: ../data/my_study.csv         # relative to this YAML file
  columns: {ID: ID, TIME: TIME, AMT: AMT, DV: DV, ROUTE: ADM}
  evid_rule: amt_present
  route_codes: {"1": iv_bolus, "2": sc}
  units: {time: h, amount: ug, concentration: ug/mL}
  molecular_weight_g_per_mol: {drug: 150000, source: "declared by analyst"}
  assay: {endpoints: {default: free_drug}}
  coincident_convention: pre_dose     # or post_dose
model:
  template: tmdd2c_full
  version: 1.1.0                     # 1.1.0 adds the SC depot
  parameters:                        # init, bounds, estimated or fixed, random effect
        (omega^2)
    CL:   {init: 0.008, lower: 1.0e-4, upper: 0.5, estimate: true,  eta: 0.3}
    kon:  {init: 1.0,   lower: 0.01,   upper: 100, estimate: false, eta: null,
```

```
            justification: "fixed: not identifiable from drug-only data"}
  residual: {model: lognormal, sd_init: 0.3}
calibration: {max_attempts: 16, n_lhs: 7, workers: 4, seed: 101, planner: rules}
estimation:  {enabled: true, method: focei, max_retries: 3, seed: 202}
diagnostics: {vpc: {enabled: true, n_sim: 200, stratify_by: ROUTE, dose_normalize: true}}
report:      {title: "My study", formats: [html, pdf]}
```

The configuration is checked against `config/schema.json` before anything runs. Unknown fields, impossible bounds, a missing molecular weight for mass units, or random effects on fixed parameters are rejected with a message that names the field.

5.3 Step 3 — Validate the data

```
kratos validate --config config\my_study.yaml --out runs\check
```

This either prints a summary and the review flags, or exits with errors that name the rows and fields to fix. Nothing is modified in your original file.

5.4 Step 4 — Run the workflow

```
kratos run --config config\my_study.yaml --run-id my_study_v1
```

Each run gets its own directory `runs\my_study_v1\`. A run directory is never overwritten; choose a new `--run-id` for a new analysis. Add `--seed 123` to change all random seeds at once.

5.5 Step 5 — Follow progress, cancel, resume

```
kratos status --run runs\my_study_v1                # stage states and progress
kratos cancel --run runs\my_study_v1                # stops at the next safe point; records
        are kept
kratos run --config config\my_study.yaml --resume runs\my_study_v1
```

A run resumes only if the data, the configuration and the Kratos code are unchanged; otherwise Kratos explains which one differs.

5.6 Step 6 — Read the results

Open `runs\my_study_v1\report\moDoc.html` (or `moDoc.pdf`). Start with **Stage status** and **Review flags**:

- `critical` flags (for example, a failed covariance step or non-convergence) must be resolved before the model is used;
- `warning` flags (high RSE, shrinkage, boundary estimates, VPC misfit) need scientific judgement;

- `info` flags describe declared conventions.

6 Command reference

Command	Purpose	Key options
`kratos validate`	check and normalise a dataset	`--config`, `--data`, `--out`
`kratos simulate`	simulate the design with the Python and/or R engine	`--engine python|r|both`, `--params`, `--out`
`kratos calibrate`	create a run and stop after calibration	`--config`, `--run-id`, `--seed`, `--planner`
`kratos run`	full workflow	`--config`, `--run-id`, `--resume`, `--seed`, `--planner rules|fixed|llm`
`kratos estimate`	nlmixr2 FOCEi on an existing run	`--run`
`kratos diagnose`	diagnostics + VPC on an existing run	`--run`
`kratos report`	rebuild figures, tables and report (no refitting)	`--run`
`kratos status` / `kratos cancel`	inspect / stop a run	`--run`
`kratos ui`	open the browser interface	`--port` (default 8501)
`kratos split`	split a dataset into two parts by ID (grouped by molecule)	`--src`, `--outdir`, `--alias A=B`
`kratos crosscheck`	predict one run's data with another run's model	`--run-a`, `--run-b`, `--out`
`kratos bootstrap`	subject-level bootstrap of a finished fit	`--run`, `--n`, `--seed`
`kratos evaluate`	adaptive vs fixed planner on replicated synthetic data	`--config`, `--replicates`, `--out`
`kratos synth`	generate the synthetic dataset and invalid-input fixtures	`--outdir`, `--seed`

Every scientific command exits with a non-zero code on failure, so it can be used in scripts. `python -m kratos ...` works the same as `kratos ...`.

7 Configuration reference

Section	Important fields	Notes
`data`	`path`, `columns`, `units`, `molecular_weight_g_per_mol`, `assay.endpoints`, `coincident_convention`, `unsorted_time_policy`, `combined_dose_observation`, `aggregate_data`	mass units need a declared molecular weight; `aggregate_data: true` marks mean-profile data
`model`	`template`, `version`, `parameters` (`init`, `lower`, `upper`, `estimate`, `eta`, `justification`), `residual`, `structure_decisions`	ksyn is always derived as kdeg·R0
`solver`	`rtol`, `atol` (Python), `r_rtol`, `r_atol` (R)	stiff solvers; failures are never hidden
`calibration`	`objective`, `optimizer`, `n_lhs`, `max_attempts`, `batch_size`, `max_nfev`, `time_budget_s`, `plateau_window`, `success_repeats`, `workers`, `planner`, `seed`	budgets stop the search; all attempts are recorded
`estimation`	`enabled`, `method: focei`, `max_retries`, `cov_method`, `n_candidates`, `timeout_s`, `seed`	FOCEi is the only enabled method
`diagnostics`	`thresholds`, `vpc` (`n_sim`, `n_bins`, `stratify_by`, `dose_normalize`), `plausibility`	thresholds are heuristics
`report`	`title`, `formats: [html, pdf]`, figure size and dpi	
`llm`	`enabled`, `model`	optional; needs `ANTHROPIC_API_KEY` and the `anthropic` package

8 The browser interface

```
kratos ui          # then open http://localhost:8501
```

- **Project setup:** upload a CSV (or give a path), map the columns, choose units, molecular weight, assay and conventions, edit parameters (tick *estimated* or leave unticked for *fixed*), validate, save the configuration (replacing an existing one needs an explicit confirmation), and launch a run.
- **Runs:** choose any previous run to see the current stage, analysis mode, calibration history, estimates, review flags and figures; download the report and export bundle; request cancellation.

Runs launched from the interface execute in a separate background process, so closing or reopening the page never restarts or stops an analysis.

9 What a run directory contains

Path	Content
`input/`	untouched copies of your CSV and configuration
`config_snapshot.json`, `provenance.json`	exact configuration used, file hashes, code hash, seeds, software versions
`status.json`, `audit.jsonl`	stage states and a time-stamped record of every action and decision
`data/`	normalised data, subject-ID map, validation report, unit conversions
`model/`	model specification and model card
`calibration/`	every attempt (including failures), ranked candidates, accepted starts, sensitivity
`estimation/final/`	nlmixr2 model code, `fit.rds`, parameters, Ω, predictions, individual parameters
`diagnostics/`	diagnostics summary, VPC simulations and statistics, bootstrap (if run)
`tables/`, `figures/`, `figures_R/`	result tables; figures as PNG (300 dpi) and SVG/PDF, with the data behind each figure
`report/`	`moDoc.html`, `moDoc.pdf`, results narrative
`export/`	a zip bundle of the run, scanned so that no secrets are included
`manifest.json`	checksums of every file

10 Reports

One run. Rebuilds figures, tables and the report from saved results in a few minutes, without refitting:

```
kratos report --run runs\my_study_v1
```

Whole project. One report covering several runs, the dataset split, the cross-part check, the planner evaluation, the tests and the limitations, written as a paper (abstract, hypothesis, aims, methods, results, discussion). Produces HTML and PDF:

```
python scripts\build_project_report.py --runs runs\synthetic_demo_v1 runs\mab_partA_v1
         runs\mab_partB_v1
start reports\project\moDoc.pdf
```

Regenerate only figures and tables, without rendering the report (Python or R):

```
python scripts\results.py --run runs\my_study_v1 --no-render
Rscript R\results.R --run runs\my_study_v1
```

The report template is `reports/moDoc.qmd` and its style is `reports/kratos.css`. After editing either, rerun one of the commands above; do not render the copies inside run folders.

11 Reproducing the mAb analysis in this project

```
cd C:\Users\devamdesai\Downloads\Kratos
python scripts\split_dataset.py --src "C:\Users\devamdesai\UF Dropbox\Devam Desai\Research\ML
        NODE-mPBPK Human PK\Data\mAb_dataset.csv" --outdir data\mab
kratos run --config config\mab_partA.yaml --run-id mab_partA_v2
kratos run --config config\mab_partB.yaml --run-id mab_partB_v2
kratos crosscheck --run-a runs\mab_partA_v2 --run-b runs\mab_partB_v2 --out reports\crosscheck
python scripts\build_project_report.py --runs runs\synthetic_demo_v1 runs\mab_partA_v2
        runs\mab_partB_v2
```

Each part takes roughly 1–1.5 hours on a 12-core PC, mostly the FOCEi fit. The split is deterministic (fixed seed), so the same profiles fall into each part every time.

12 Advanced use

- **Change the planner:** `--planner fixed` runs the non-adaptive multistart baseline; `--planner llm` (with `llm.enabled: true` and an API key) lets an LLM propose starting values. Its proposals are schema-checked and must stay inside the bounds; otherwise the rules planner is used.
- **Bootstrap:** `kratos bootstrap --run runs\X --n 50` refits resampled subjects in 4 parallel R workers. Allow about 5–25 minutes per replicate.
- **Hypothesis evaluation:** `kratos evaluate --config config\synthetic_demo.yaml --replicates 5 --out evaluation\my_eval`.
- **Cross-engine check:** `kratos simulate --config config\synthetic_demo.yaml --engine both --out sim.csv` compares Python and R predictions against the prespecified tolerance.

13 Troubleshooting

Symptom	Cause and fix
`CONFIG ERROR ... schema validation`	a field name or value is not allowed; the message names the field. Compare with `config/schema.json`.
`Data validation failed: ... [data rows ...]`	fix the listed rows or declare the convention (e.g. `unsorted_time_policy: sort_and_flag`, `combined_dose_observation: split`).
`UnsupportedDataError ... censored`	censored (BLQ) records are not modelled yet; remove the censoring flag only by an explicit, documented decision.
Estimation stage `not_run: R backend unavailable`	install R and the packages (`R\setup_packages.R --install`) or set `KRATOS_RSCRIPT`.
`stringfish.dll ... procedure could not be found`	a package binary built for a newer R; run `R\setup_packages.R --install`, which places a compatible RcppParallel in `.Rlib`.
`calibration_failed`	no attempt converged; increase `max_nfev` or `max_attempts`, or check bounds and initial values (see `calibration_history.csv`).
Covariance step failed / parameter at bound	the parameter is not identifiable from the data; fix it or reparameterise (a scientist decision).
PDF not produced	install Microsoft Edge or Chrome, or set `KRATOS_BROWSER` to the browser executable.
Unicode error in the console on Windows	run `$env:PYTHONIOENCODING = "utf-8"` first.
`FileExistsError` when starting a run	the run id already exists; choose a new `--run-id`.

14 Glossary

Term	Meaning
TMDD	target-mediated drug disposition: drug binding to its pharmacological target affects its pharmacokinetics
Calibration	pooled least-squares fit used only to find good starting values
FOCEi	first-order conditional estimation with interaction (nlmixr2), the population estimation method
OFV	objective function value of the population fit (not comparable with the calibration objective)
BSV / Ω	between-subject (here between-profile) variability of parameters
Shrinkage	how much individual random-effect estimates are pulled toward zero; high values weaken EBE-based diagnostics
CWRES	conditional weighted residuals
VPC	visual predictive check: observed percentiles compared with simulated ones
Review flag	heuristic warning for the scientist (critical / warning / info), separate from execution status